

Train Less, Contribute More: A Neural-Architecture Search for Numerai’s Ender Target

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Abstract

We search for the neural network that best predicts Numerai’s 20-day Ender target, judging models not by raw correlation but by benchmark model contribution (BMC): the signal a model adds beyond Numerai’s own LightGBM benchmark. Across 35 cross-validated runs in six rounds we compare plain, wide, deep and residual multilayer perceptrons (MLPs), a TabM-style BatchEnsemble, learned per-bin feature embeddings, two loss functions and the main training hyperparameters. The best model is a plain three-layer MLP (1024–512–256) trained with a per-era Pearson-correlation loss for a short time. Averaged over three seeds it reaches BMC 0.0089 over the last 200 out-of-fold eras, $2.1\times$ a LightGBM baseline trained on the same data. Architecture mattered far less than one variable: the amount of training. Training moves predictions steadily away from the benchmark, raising BMC, until it starts to destroy raw signal; every knob we varied (epochs, learning rate, batch size, dropout) lands on this single curve. On 613 eras never used during the search, the result transfers once the training amount is held fixed in optimizer steps rather than epochs: the 6-epoch recipe over-trains on twice the data, while 4 epochs recovers BMC 0.0082. This matches the continuous-time view of early stopping as ridge regularization.

1 Introduction

Numerai runs a stock-market prediction tournament on obfuscated data: each week (an *era*) participants rank several thousand stocks from a few thousand anonymized features, and are paid for predictions that improve Numerai’s meta model. A model with high raw correlation that merely re-discovers what Numerai’s benchmark models already know contributes little. Benchmark model contribution (BMC) measures this directly as the correlation of a model’s predictions with the target after neutralizing them against the benchmark’s predictions [47, 50].

Gradient-boosted trees dominate tabular prediction [26], and the tournament’s benchmark is itself a LightGBM model. Neural networks therefore have a natural advantage for BMC: their inductive bias differs from that of trees, so even at lower raw correlation they may carry more *unique* signal. This paper asks which neural architecture, loss and training recipe maximize that unique signal.

Our contributions are:

1. A controlled architecture search, one variable at a time with seed-averaged confirmation, showing that a plain MLP beats residual, ensemble and embedding-based alternatives on BMC.
2. Evidence that BMC is governed by the *amount* of training, and that this trade-off has a single interior optimum shared by all training knobs.
3. A confirmation on disjoint data showing that the optimum is a fixed number of optimizer steps, so the epoch count must shrink as the training set grows.

2 Related work

Machine learning for cross-sectional returns. Neural networks and trees improve out-of-sample return prediction over linear models through nonlinear interactions, and shallow networks outperform deeper ones [29]; later work learns factor structure [30] or a stochastic discount factor with moment-based losses [16], and finds machine-learning profits concentrated in hard-to-arbitrage stocks [3]. Whether model complexity is a virtue [36] remains debated [20, 49]. Losses closer to the trading objective, such as ranking [44, 51], classification [5], Sharpe-ratio [64] and correlation losses [48], often beat squared error. Hedging unpriced risk [18] motivates the feature neutralization common on Numerai [11, 13]; forecast combination [53], auxiliary targets [46], dynamic ensembling under regime change [61] and era boosting [54] are other recurring tools. Repeated configuration search inflates backtests [2, 6, 7]; shrinkage is a classic remedy [39].

Tabular deep learning. Recent work revisits MLP, ResNet and transformer baselines [23], numerical feature embeddings [24], parameter-efficient ensembles [25], well-tuned default MLPs [33] and heavily regularized plain networks [35]. Attention-based models [58] and tabular foundation models [27, 28, 32] do not yet scale to millions of rows with hundreds of features. Weight averaging [34], deep ensembles [40] and model soups [62] reduce variance.

Training dynamics and differential equations. Gradient flow stopped at time t behaves like ridge regression with penalty $\lambda = 1/t$ [1]; finite learning rates add an implicit gradient penalty whose strength scales with the learning rate and, for stochastic gradients, inversely with batch size [8, 41, 57]. Decay towards the initialization is the explicit counterpart of early stopping [43], and noise injection acts as Tikhonov or heat-equation smoothing [9, 14, 15, 59]. Continuous-depth networks [17, 19, 21, 31, 37, 56] and their stochastic and controlled variants [38, 42, 45] have been applied to finance [22, 63], building on stochastic-calculus models of prices [10], but we found no application to static cross-sectional ranking.

Number theory. Its one documented use relevant here is quasi-Monte Carlo search: low-discrepancy sequences cover a hyperparameter space more evenly than random draws [12, 55]. Random Fourier features [4, 52] and feature hashing [60] are related tools that do not fit dense, low-cardinality features.

3 Experimental setup

Data. Numerai v5.3, `medium` feature set (780 features, each an integer bin 0–4), target `target_ender_20`. Because experiments ran on a 16 GB laptop, the search used every fourth era of the training and validation data (307 eras, 1.7M rows; *scout data*). The confirmation step used every second era at offset one (613 eras, 3.4M rows; *scale data*), which shares no eras with the scout data.

Validation. Five expanding-window folds over eras with a 13-era embargo. The first fold has no training data, so out-of-fold (OOF) predictions cover the last four folds. Every network trains for a fixed number of epochs; there is no early stopping, so no validation information leaks into OOF scores.

Metrics. The primary metric is mean BMC over the last 200 OOF eras against the official `v53_lgbm_ender20` predictions. We also report BMC over all OOF eras, raw per-era correlation (`corr`), and the mean correlation between a model’s predictions and the benchmark’s.

Architecture (MSE, 4 epochs)	corr	BMC	BMC last 200	corr w/ benchmark
Wide MLP 1024–512–256	0.0259	0.0061	0.0062	0.404
MLP 256–128	0.0249	0.0053	0.0052	0.402
LightGBM baseline	0.0277	0.0042	0.0042	0.485
Residual MLP (2 blocks, width 256)	0.0187	0.0036	0.0037	0.319
TabM BatchEnsemble ($k = 8$)	0.0213	0.0026	0.0026	0.393
MLP with per-bin embeddings	0.0157	0.0012	0.0006	0.312

Table 1: Round-1 architecture scout on scout data, single seeds.

Models. All networks share one PyTorch implementation trained on Apple-silicon GPUs. Inputs are scaled to $(x - 2)/2$ unless embeddings are used. The MSE loss uses random batches of 4,096 rows. The correlation loss uses batches of four whole eras and minimizes the negative mean per-era Pearson correlation

$$\mathcal{L}_{\text{corr}} = -\frac{1}{|E|} \sum_{e \in E} \frac{\langle \hat{y}_e - \bar{\hat{y}}, y_e - \bar{y} \rangle}{\|\hat{y}_e - \bar{\hat{y}}\| \|y_e - \bar{y}\|}. \quad (1)$$

Optimization uses AdamW with a cosine schedule and 5% warm-up. The baseline is LightGBM with 2,000 trees and 31 leaves on identical data, features and folds.

Protocol. Rounds of four to eight configurations, each changing one variable from the current leader. Gaps smaller than seed noise were re-checked with three-seed averages before being trusted. The search stopped after two consecutive rounds without a confirmed improvement.

4 Results

4.1 A plain MLP beats more elaborate architectures

Under an identical recipe (MSE, four epochs) the plain MLP beats LightGBM on BMC despite lower raw correlation (Table 1, Figure 1). Its predictions correlate only 0.40 with the benchmark, against 0.49 for LightGBM, so more of its signal is new. Every attempt to add structure did worse. Learned per-bin embeddings quadruple the input width and overfit; with only five bin values a piecewise-linear embedding [24] reduces to one-hot encoding. Residual blocks and TabM ensembling also lost, as did a deeper (512–256–128) and a wider (2048–1024–512) network in later rounds.

4.2 Training amount governs contribution

Training directly on per-era correlation lowers the benchmark correlation of predictions far more than MSE does, but at 20 epochs it overfits badly: training-era correlation reached 0.24 against 0.018 out of fold. More dropout and weight decay, or an added MSE term, did not fix this as well as simply training less.

Figure 2 shows the mechanism. As the network trains, its predictions drift steadily away from the benchmark (panel b), which raises BMC, while raw correlation holds roughly flat and then falls. BMC therefore peaks just before raw correlation collapses (panel a). Every knob that changes the amount of training lands on this curve: the optimal epoch count moves from 8 to 6 when the learning rate doubles; a learning rate of 5×10^{-4} under-trains while 3×10^{-3} over-trains; eight eras per batch halves the number of updates and under-trains; and extra dropout or input dropout at the optimal length behaves like less training, pulling predictions back towards the benchmark. Across all scout runs, BMC rises as benchmark correlation falls until raw signal is lost (Figure 3).

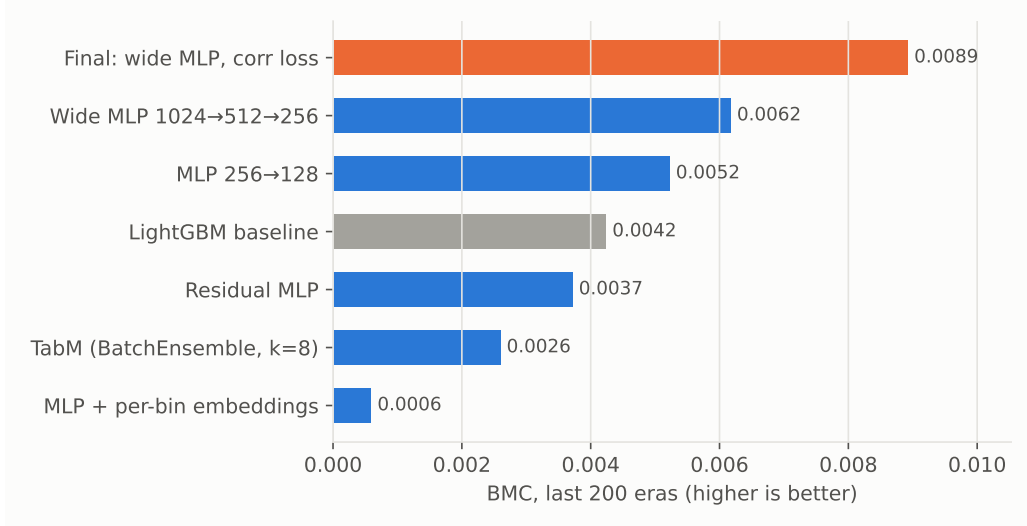


Figure 1: BMC over the last 200 OOF eras. Blue bars share the untuned round-1 recipe; gray is the LightGBM baseline; orange is the final tuned model (correlation loss, learning rate 2×10^{-3} , six epochs, three-seed average).

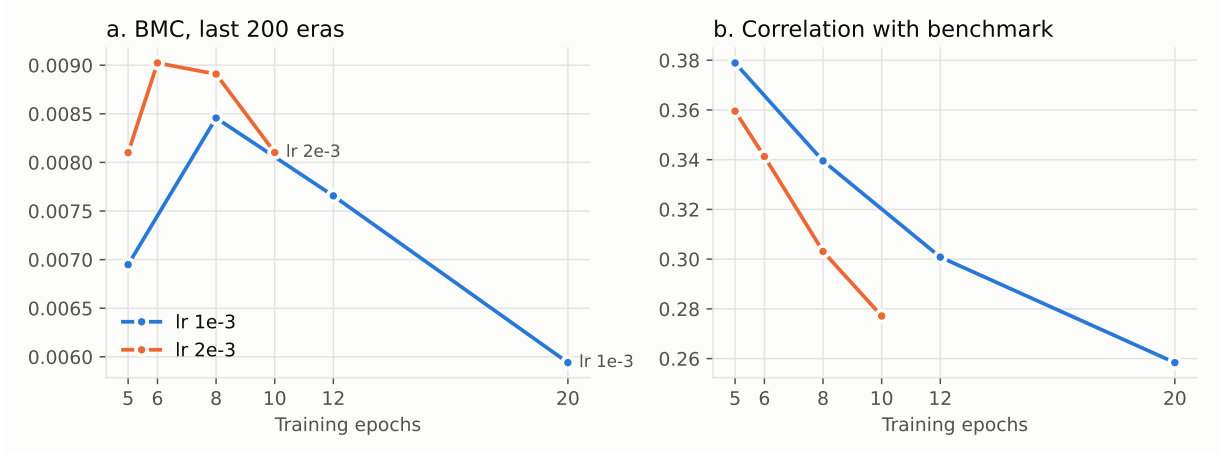


Figure 2: Correlation-loss MLP (256–128), single seeds. (a) BMC peaks at a training length that depends on the learning rate. (b) Correlation with the benchmark falls the longer the network trains. The point at learning rate 2×10^{-3} and eight epochs is a lucky seed; its three-seed average is 0.0083.

4.3 Seed noise and the confirmed winner

Single-seed results vary by up to ± 0.0006 BMC: one run at learning rate 2×10^{-3} scored 0.0089, but its three-seed average was 0.0083. We therefore judged finalists on three-seed averages (Table 2). The wide and narrow winners’ predictions correlate 0.96, so averaging them adds nothing. Width helps only once training is short enough to stop the network from memorizing. Figure 4 shows that the winner’s advantage accumulates steadily across eras rather than coming from a few outliers.

4.4 Confirmation on unseen eras

We retrained the finalists on the scale data: 613 eras never used during the search, with twice the training rows per fold (Table 3). Run unchanged, the six-epoch winner dropped to 0.0056, with exactly the signature of over-training seen in Figure 2: lower raw correlation and lower benchmark correlation (0.28 against 0.34 on scout data). With twice the rows per epoch, six epochs means twice as many optimizer steps. Cutting to four epochs recovers BMC 0.0082 and

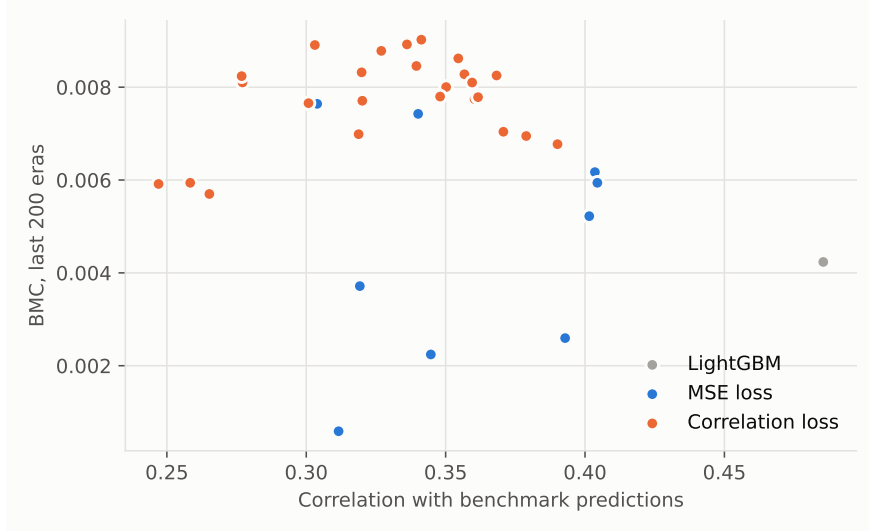


Figure 3: All scout runs: correlation with the benchmark against BMC. Lower benchmark correlation buys contribution until raw signal collapses.

Model (scout data, 3 seeds)	corr	BMC	BMC last 200	Sharpe	w/ bench.
Wide 1024–512–256, corr loss, lr 2e-3, 6 ep.	0.0260	0.0088	0.0089	0.71	0.336
Narrow 256–128, corr loss, lr 2e-3, 6 ep.	0.0262	0.0085	0.0086	0.71	0.355
Narrow 256–128, corr loss, lr 1e-3, 8 ep.	0.0261	0.0082	0.0083	0.68	0.357
Rank average of wide and narrow	0.0263	0.0087	0.0088	0.71	–

Table 2: Seed-averaged finalists on scout data.

returns benchmark correlation to its scout level (0.33), on data the search never touched. The optimal training amount is therefore a fixed number of optimizer steps, not epochs, consistent with the continuous-time view in which stopping time sets the effective regularization [1].

5 Discussion

Why does less capacity win? Numerai’s per-era signal is weak, with correlations around 0.02, so spare capacity is spent fitting era-specific noise. The architectures that lost all add capacity or flexibility; the winning changes all limit how far the network moves from its initialization. This mirrors the shallow-network result for U.S. equities [29] and the strength of regularized plain MLPs on tabular data [35].

Why does a correlation loss help bmc? Squared error rewards predicting the target’s overall level and spread, which a tree benchmark already does well. A per-era Pearson loss rewards only within-era ranking, the quantity Numerai scores, and in practice yields predictions far less similar to the benchmark’s.

A practical recipe. For a new target or feature set, tune the amount of training first, in optimizer steps, and pick the BMC peak before adding architecture. When the training set grows, scale the epoch count down so the number of steps stays at the optimum.

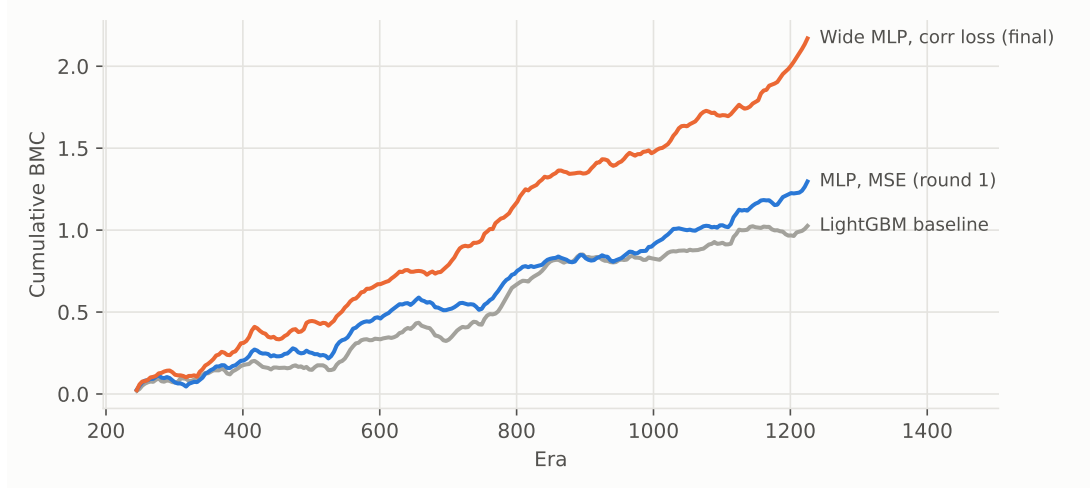


Figure 4: Cumulative per-era BMC on the same 246 scout OOF eras.

Model (3 seeds)	corr	BMC	BMC last 200	corr w/ benchmark
Wide, 6 epochs, scout data (reference)	0.0260	0.0088	0.0089	0.336
Wide, 4 epochs, scale data	0.0237	0.0074	0.0082	0.330
Narrow, 6 epochs, scale data	0.0220	0.0068	0.0067	0.308
Wide, 6 epochs, scale data	0.0206	0.0065	0.0056	0.283

Table 3: Confirmation on 613 eras disjoint from the search. The last 200 eras of the scale data are more recent than those of the scout data, so absolute levels are not directly comparable.

Limitations. We used the `medium` feature set only; the 3,555-feature set did not fit in memory. Since Numerai’s round 1343 the official BMC uses the 60-day Ender target and a stake-weighted ensemble of benchmark models [50], so our 20-day, single-benchmark BMC is a proxy for the payout metric. Finalists were averaged over three seeds, so differences below about 0.0003 should be read as ties. Thirty-five configurations were tried on the same folds; a deflated Sharpe ratio [6] would temper the best scout result, which is one reason we confirmed on disjoint eras. No live-tournament results are included yet.

6 Conclusion

On Numerai’s 20-day Ender target the best neural network is not an exotic one. A plain three-layer MLP with a per-era correlation loss, trained briefly, doubles the LightGBM baseline’s benchmark contribution. Architecture mattered less than the amount of training, which trades raw signal for uniqueness along a single curve, and that amount is best fixed in optimizer steps. Next steps follow directly from the literature: interpolating trained weights towards their initialization as a continuous training-amount dial [1], penalizing correlation with the benchmark during training, training on the 60-day target that Numerai now pays on, and a quasi-Monte Carlo search over the learning-rate and step-count interaction [12].

Reproducibility. Code, configurations, figures and this paper are in <https://github.com/VolodymyrLinuxovich/example-scripts>, branch `nn-arch-ender20`, directory `numerai/agents/experiments/nn_arch_ender20`.

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A Full experiment log

Every configuration that ran, by round. Single seeds unless the configuration says otherwise; bold marks each round’s best BMC over the last 200 eras.

Baselines

Run	Configuration	corr	BMC	last 200	w/ bench.
r0_baseline_small_lgbm	LightGBM, 2,000 trees	0.0277	0.0042	0.0042	0.485
r0_smoke_mlp	MLP 256–128 · MSE · 1 ep · lr 0.001	0.0188	0.0024	0.0022	0.345

Round 1 · architecture

Run	Configuration	corr	BMC	last 200	w/ bench.
r1_mlp_wide_mse	MLP 1024–512–256 · MSE · 4 ep · lr 0.001 · drop 0.2	0.0259	0.0061	0.0062	0.404
r1_mlp_corr	MLP 256–128 · corr loss · 20 ep · lr 0.001	0.0184	0.0055	0.0059	0.258
r1_mlp_mse	MLP 256–128 · MSE · 4 ep · lr 0.001	0.0249	0.0053	0.0052	0.402
r1_resmlp_mse	Residual MLP w256 · MSE · 4 ep · lr 0.001	0.0187	0.0036	0.0037	0.319
r1_tabm_mse	TabM 256–256 · MSE · 4 ep · lr 0.001	0.0213	0.0026	0.0026	0.393
r1_mlp_embed_mse	MLP 256–128 · MSE · 4 ep · lr 0.001 · embeddings	0.0157	0.0012	0.0006	0.312

Round 2 · training length

Run	Configuration	corr	BMC	last 200	w/ bench.
r2_corr_ep8	MLP 256–128 · corr loss · 8 ep · lr 0.001	0.0253	0.0083	0.0085	0.340
r2_corr_epb1	MLP 256–128 · corr loss · 5 ep · lr 0.001 · 1 era/batch	0.0254	0.0078	0.0080	0.350
r2_wide_ep8	MLP 1024–512–256 · MSE · 8 ep · lr 0.001 · drop 0.2	0.0229	0.0073	0.0076	0.304
r2_mse_ep8	MLP 256–128 · MSE · 8 ep · lr 0.001	0.0244	0.0071	0.0074	0.340
r2_corr_reg	MLP 256–128 · corr loss · 20 ep · lr 0.001 · drop 0.3 · wd 0.001	0.0230	0.0068	0.0070	0.319
r2_wider_mse	MLP 2048–1024–512 · MSE · 4 ep · lr 0.001 · drop 0.3	0.0255	0.0057	0.0059	0.404
r2_wide_corr	MLP 1024–512–256 · corr loss · 20 ep · lr 0.001 · drop 0.2	0.0183	0.0059	0.0059	0.247
r2_corr_mse	MLP 256–128 · corr+MSE · 20 ep · lr 0.001	0.0184	0.0052	0.0057	0.265

Round 3 · refine

Run	Configuration	corr	BMC	last 200	w/ bench.
r3_corr_ep8_seeds3	MLP 256–128 · corr loss · 8 ep · lr 0.001 · 3 seeds	0.0261	0.0082	0.0083	0.357
r3_wide_corr_ep6	MLP 1024–512–256 · corr loss · 6 ep · lr 0.001 · drop 0.2	0.0261	0.0079	0.0077	0.360
r3_corr_ep12	MLP 256–128 · corr loss · 12 ep · lr 0.001	0.0225	0.0073	0.0077	0.301
r3_corr_ep5	MLP 256–128 · corr loss · 5 ep · lr 0.001	0.0253	0.0067	0.0069	0.379
r3_corr_ep8_drop03	MLP 256–128 · corr loss · 8 ep · lr 0.001 · drop 0.3	0.0260	0.0067	0.0068	0.390

Round 4 · new knobs

Run	Configuration	corr	BMC	last 200	w/ bench.
r4_lr2e3	MLP 256–128 · corr loss · 8 ep · lr 0.002	0.0237	0.0084	0.0089	0.303
r4_indrop01	MLP 256–128 · corr loss · 8 ep · lr 0.001 · in-drop 0.1	0.0263	0.0079	0.0083	0.368
r4_epb8	MLP 256–128 · corr loss · 8 ep · lr 0.001 · 8 era/batch	0.0250	0.0072	0.0078	0.362
r4_deep	MLP 512–256–128 · corr loss · 8 ep · lr 0.001	0.0241	0.0076	0.0077	0.320
r4_lr5e4	MLP 256–128 · corr loss · 8 ep · lr 0.0005	0.0252	0.0068	0.0070	0.371

Round 5 · lr 2e-3

Run	Configuration	corr	BMC	last 200	w/ bench.
r5_lr2e3_ep6	MLP 256–128 · corr loss · 6 ep · lr 0.002	0.0257	0.0087	0.0090	0.341
r5_wide_lr2e3_ep6	MLP 1024–512–256 · corr loss · 6 ep · lr 0.002 · drop 0.2	0.0256	0.0088	0.0088	0.327
r5_lr2e3_seeds3	MLP 256–128 · corr loss · 8 ep · lr 0.002 · 3 seeds	0.0242	0.0080	0.0083	0.320
r5_lr3e3	MLP 256–128 · corr loss · 8 ep · lr 0.003	0.0217	0.0078	0.0082	0.277
r5_lr2e3_ep10	MLP 256–128 · corr loss · 10 ep · lr 0.002	0.0216	0.0075	0.0081	0.277

Round 6 · seed check

Run	Configuration	corr	BMC	last 200	w/ bench.
r6_wide_lr2e3_ep6_seeds3	MLP 1024–512–256 · corr loss · 6 ep · lr 0.002 · drop 0.2 · 3 seeds	0.0260	0.0088	0.0089	0.336
r6_lr2e3_ep6_seeds3	MLP 256–128 · corr loss · 6 ep · lr 0.002 · 3 seeds	0.0262	0.0085	0.0086	0.355
r6_lr2e3_ep5	MLP 256–128 · corr loss · 5 ep · lr 0.002	0.0258	0.0080	0.0081	0.360
r6_wide_lr2e3_ep5	MLP 1024–512–256 · corr loss · 5 ep · lr 0.002 · drop 0.2	0.0257	0.0079	0.0078	0.348

Scale · unseen eras

Run	Configuration	corr	BMC	last 200	w/ bench.
s1_wide_ep4_seeds3	MLP 1024–512–256 · corr loss · 4 ep · lr 0.002 · drop 0.2 · 3 seeds	0.0237	0.0074	0.0082	0.330
s1_narrow_ep6_seeds3	MLP 256–128 · corr loss · 6 ep · lr 0.002 · 3 seeds	0.0220	0.0068	0.0067	0.308
s1_wide_ep6_seeds3	MLP 1024–512–256 · corr loss · 6 ep · lr 0.002 · drop 0.2 · 3 seeds	0.0206	0.0065	0.0056	0.283